

Jonathan Cruz

SENIOR SOFTWARE ENGINEER · RISK & COMPLIANCE SYSTEMS

📍 SÃO PAULO, BRAZIL ✉ JONATHAN.WILLIAN.TOD@GMAIL.COM 🌐 AKAMURAASAI 🔗 AKAMURAASAI.WORK

PROFILE

"Senior software engineer with 10+ years building fintech and payments platforms with strong emphasis on risk, compliance, and transactional security. Deep hands-on experience with KYC/EDD systems, UBO verification, FinCEN/CTA compliance, idempotency, and audit-friendly event-driven architectures. Strong in microservices on Postgres + cloud (AWS, GCP), CI/CD automation, TDD/BDD, and platform-level patterns that scale across product teams. Mentor and tech lead – comfortable taking ownership of medium-to-large components end-to-end and shipping under hyper-growth conditions."

PROFESSIONAL EXPERIENCE

SELECTED ROLES & CONTRIBUTIONS

FINTech FOR FUND
ADMINISTRATION

Sydecar

FULL-STACK SOFTWARE
ENGINEER

KYC/EDD

ANTI-FRAUD &
COMPLIANCE SYSTEMS

Designed and implemented the Enhanced Due Diligence system end-to-end: LP funding blocks, UBO (Ultimate Beneficial Owner) verification, FinCEN integration, CTA compliance, and automatic KYC revalidation for active investors. Built race-resistant verify-identity flow with re-check inside a to handle "email-free-at-blur → taken-at-submit" edge case (409 EMAIL_NOW_EXISTS).

KYC EDD FINCEN ANTI-FRAUD

Transactional
Security

IDEMPOTENCY AT
SCALE

Implemented webhook deduplication via (unique) with a PROCESSING/PROCESSED/REJECTED state machine, and an API-level idempotency interceptor with 48h TTL scoped by . Patterns directly applicable to payment fraud detection and transactional integrity at scale.

IDEMPOTENCY WEBHOOKS TRANSACTIONAL SECURITY

Event-
driven
Architecture

BULLMQ + OUTBOX

Standardized event delivery using BullMQ Pro with exponential backoff (5 retries / 3s base in ledger; 20 retries / 10s base in pubsub). Outbox pattern: persist resolution first, enqueue after — jobs idempotent and survive enqueue failure. Global per-deal ordering via Redis INCR counters with deterministic predecessor lookup, enabling auditable reconstructable event chains.

EVENT-DRIVEN BULLMQ OUTBOX AUDIT

Engineering-
led
Ownership

ADRS

Drove Billing V2 design and delivery: wrote the ADR, aligned cross-functional stakeholders (compliance, finance, product), defined the quarterly fee scheduler and billing model versioning, then shipped. Same ownership pattern applied to ts-rest migration, KYC revalidation, and webhook standardization.

ADRS CROSS-FUNCTIONAL TECH LEAD

External
Integrations

TDD + BDD
WORKFLOWS

Built bidirectional QuickBooks invoice sync with mock service for local dev, state validation, auto-sync on approval, and full E2E suite using Playwright and Cucumber BDD. Worked on Plaid Link integration with processor tokens powering ACH via Treasury Prime — including mock banking engine for local sandbox.

CUCUMBER BDD PLAYWRIGHT PLAID TREASURY PRIME

Type-safe
Microservices

NESTJS MONOREPO

Migrated legacy controllers (auth, onboarding, admin) to type-safe ts-rest contracts with Zod schemas across an Nx monorepo of dozens of NestJS libraries, eliminating contract-mismatch bugs at scale. Built and deployed cloud microservices integrated with Postgres (Prisma), Redis, and external SaaS providers.

NESTJS PRISMA POSTGRES REDIS

MAY 2023 — PRESENT
SAN FRANCISCO, CA (REMOTE)

B2B PAYMENTS PLATFORM

Zax

SR SOFTWARE ENGINEER

JAN 2019 — MAY 2023
SÃO PAULO

B2B Payment Flows

CREDIT RISK
VALIDATION

Built from scratch a wholesale ordering platform with integrated payment flows and per-buyer credit-limit validation, applied during order-to-cash to prevent over-extension. Built with React Native, TypeScript, and GraphQL.

REACT NATIVE TYPESCRIPT GRAPHQL CREDIT RISK

Cloud Infrastructure

AWS OPERATIONS

Deployed and operated full cloud infrastructure on AWS (EC2, ECS, S3, Lambda, VPCs) supporting high-availability B2B workloads. Containerized with Docker; built serverless React/Next.js apps. Owned the operational health of production deployments.

AWS ECS LAMBDA DOCKER

CI/CD

AUTOMATED DELIVERY

Configured end-to-end pipeline with GitHub Actions, Fastlane, and Code Climate for automated quality gates. Treated release-time toil as an engineering problem — reduced release cycle from weeks to days.

GITHUB ACTIONS FASTLANE CI/CD

CRYPTO CONSUMER FINTECH

Netlolo

SR SOFTWARE ENGINEER

SEP 2018 — JAN 2019
SÃO PAULO

Crypto Wallet

WEB3 INTEGRATION

Built a mobile and web product bringing cryptocurrency to mainstream users with React Native and TypeScript, integrating Web3 smart contracts on Ethereum. Same fraud/transactional-security mindset as traditional payments — protecting on-chain transactions and self-custodial keys.

REACT NATIVE TYPESCRIPT WEB3 ETHEREUM

Real-time SDK

HIGH-AVAILABILITY
MESSAGING

Developed a chat SDK for businesses using React Native and React/Redux, powered by XMPP and a Java backend for low-latency real-time messaging across multiple client integrations.

REACT REDUX XMPP JAVA

Zero

DOWNTIME

Built CI/CD automations ensuring continuous deployment with no service interruption. PostgreSQL with Hasura as the GraphQL layer.

POSTGRESQL HASURA CI/CD

CONSUMER MICROSERVICES

Escale

SR SOFTWARE ENGINEER

JAN 2018 — SEP 2018
SÃO PAULO

Microservices Architecture

MULTI-DB OPERATIONS

Built the product foundation with React, Redux, and a Node.js/TypeScript backend across multiple microservices. Selected MySQL, PostgreSQL, or MongoDB per service responsibility and drove DB design decisions across product teams.

MICROSERVICES MYSQL POSTGRESQL MONGODB

Mentorship

TECH TALKS

Led internal Tech Talks on RESTful design, Functional Programming, and Microservices architecture, coaching engineers across product teams. Established service-boundary and API-contract patterns adopted org-wide. Mentor mindset carried into every team since.

MENTORSHIP TECH TALKS CIRCLECI

LENDING FINTECH

PraValer

PL SOFTWARE ENGINEER

JUN 2017 — JAN 2018
SÃO PAULO

Loan Origination

LEGACY
MODERNIZATION

Migrated legacy PHP origination platform to a Node.js/Express backend for a student loans product with governmental funding. Converted manual, error-prone processes into automated, auditable loan request lifecycle, eligibility, and disbursement flows.

NODE.JS EXPRESS LOAN ORIGATION

Data Layer

PERFORMANCE TUNING

Restructured PostgreSQL transactional tables with improved schemas and indexes, eliminating N+1 queries and reducing query times by an order of magnitude. MongoDB for request logs and unstructured data — early experience with data pipelines and processing.

POSTGRESQL MONGODB QUERY OPTIMIZATION

CORE SKILLS

RISK & COMPLIANCE

KYC/EDD, UBO verification, FinCEN, CTA compliance, Idempotency, Webhook deduplication, Audit trails, Transactional security

BACKEND

TypeScript, NestJS, Node.js, Prisma, PostgreSQL, ts-rest, Zod, GraphQL, REST APIs, Microservices

EVENT-DRIVEN & DISTRIBUTED

BullMQ, Outbox pattern, Redis, Event ordering, State machines

DATA & DATABASES

PostgreSQL (deep), MySQL, MongoDB, SQL, Query optimization, Schema design

CLOUD & DEVOPS

AWS (EC2/ECS/S3/Lambda/VPC), GCP, Docker, CI/CD (GitHub Actions, Fastlane, CircleCI), TDD, Cucumber BDD, Playwright

LEADERSHIP

ADRs, Engineering-led ownership, Cross-functional collaboration, Mentorship, Tech Talks, Interview panels

LANGUAGES

PORTUGUESE

Native

ENGLISH

C1/C2

EDUCATION

Anhanguera

BACHELOR OF TECHNOLOGY (B.TECH.) IN INFORMATION TECHNOLOGY ·
JAN 2016 — JUL 2018